Fractals

What is fractal dimensions?

$m = 4$

Scaled $x \to 2x$
$y \to 2y$

$r = 2$

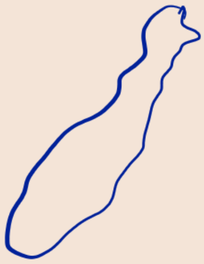
$$d = \frac{\ln m}{\ln r} = 2$$

Box dimensions

$N(\epsilon)$ scales as ϵ^{-d}

$$N(\epsilon) \sim \frac{1}{\epsilon^{d}}$$ [Haussdorff dimension]

$d \to$ dimension.



← terrible map of England.

The coastlines form fractals.

Then put it on graph paper with box edge width ϵ. Then the Number of boxes is $N(\epsilon)$.

Cantor Set

0 ———— 1 $\quad C_0 \quad L_0 = 1$

0 $\frac{1}{2}$ —— —— $\quad C_1 \quad L_1 = \frac{2}{3}$

\- \- \- \- $\quad C_2 \quad L_2 = \left(\frac{2}{3}\right)^2$

$$L_n = \left(\frac{2}{3}\right)^n$$

$$N(\epsilon) = 2^n$$

where $\epsilon \sim \left(\frac{1}{3}\right)^n$

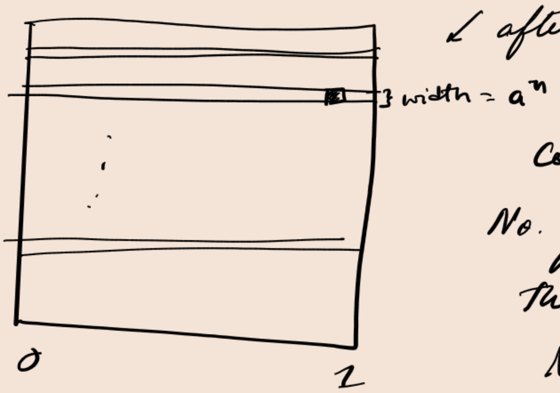
$$d = \frac{\ln 2}{\ln 3}$$

$$d = \frac{\ln 2}{\ln 3}$$ Read Strogatz.

\# Van Koch Curve.
\# Cantor Dust } Read Strogatz.
\# Serpienski Triangular.

Baker map

$$(x_{n+1}, y_{n+1}) = \begin{cases} 2x_n, \; a y_n & 0 \le x_n, y_n \le 1/2 \\ 2x_n - 1, \; a y_n + 1/2 & 1/2 \le x_n, y_n \le 1. \end{cases}$$



after some iterati. (say n iterations)

Consider box width $\epsilon = a^n$

No. of boxes in 1 strip, N_s

Thus the with of each strip

$N_s \epsilon^2 = a^n$

$N_s \sim a^{-n}$

(This is for one strip.

Total no. of strips is 2^n

$N_s \sim \dfrac{2^n}{a^n}$